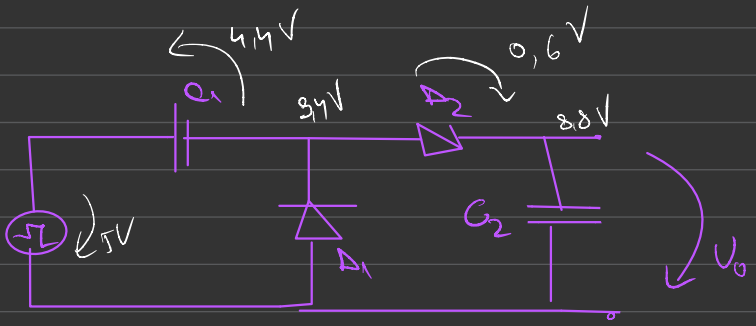
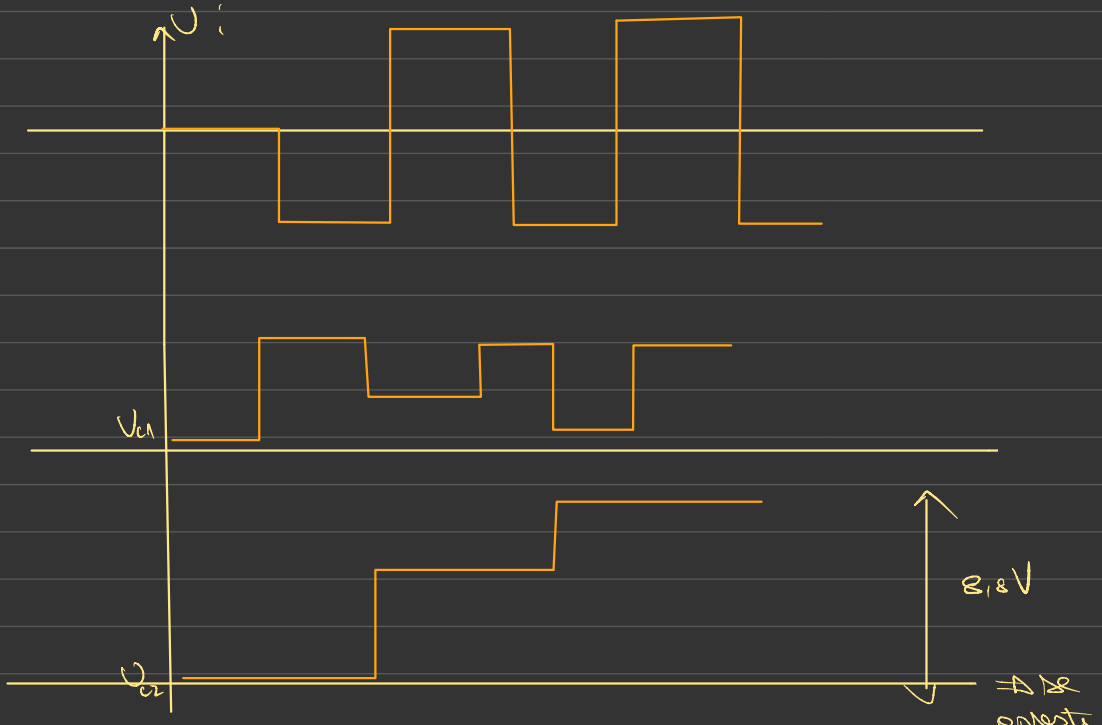
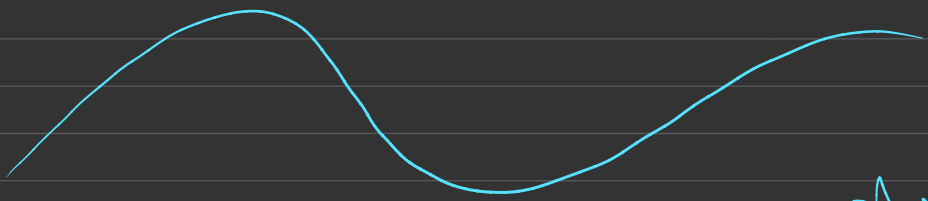
Cours 5



$\frac{1}{T}\int i_c \rightarrow V_c$

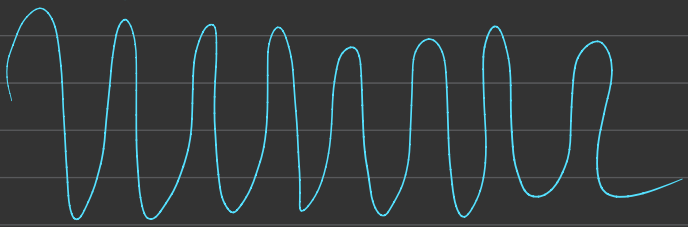
$I_c = C\frac{dV_c}{dt}$: $I_c > 0$ → $V_c \nearrow$; $I_c = 0$ → V_c ct ; $I_c < 0$ → $V_c \searrow$





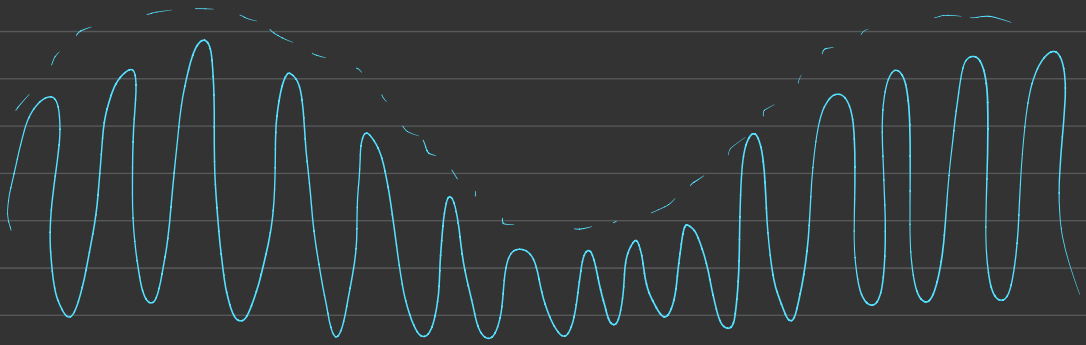
Audio
20 Hz - 20 KHz

⊗ înmulțire



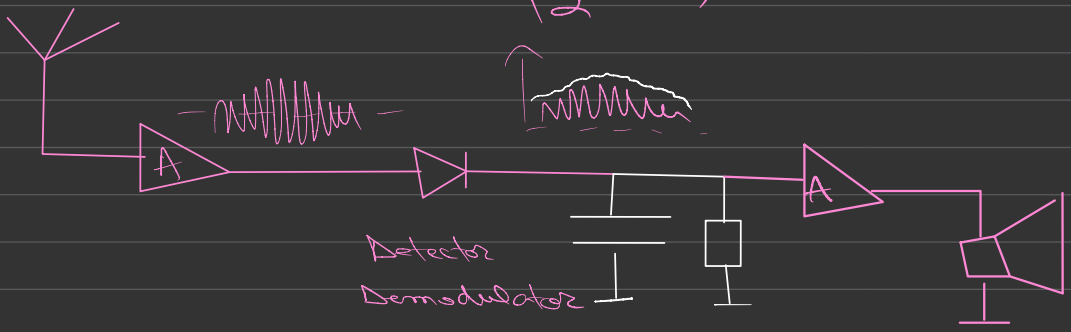
Purtătoare
800 KHz

=

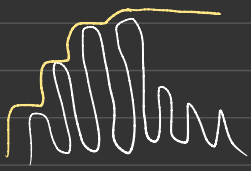


Modulație în amplitudine
(AM)
→ PAM

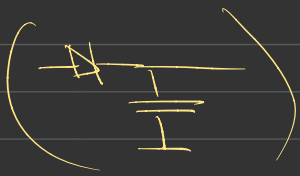
(doar parțial pozitiva)



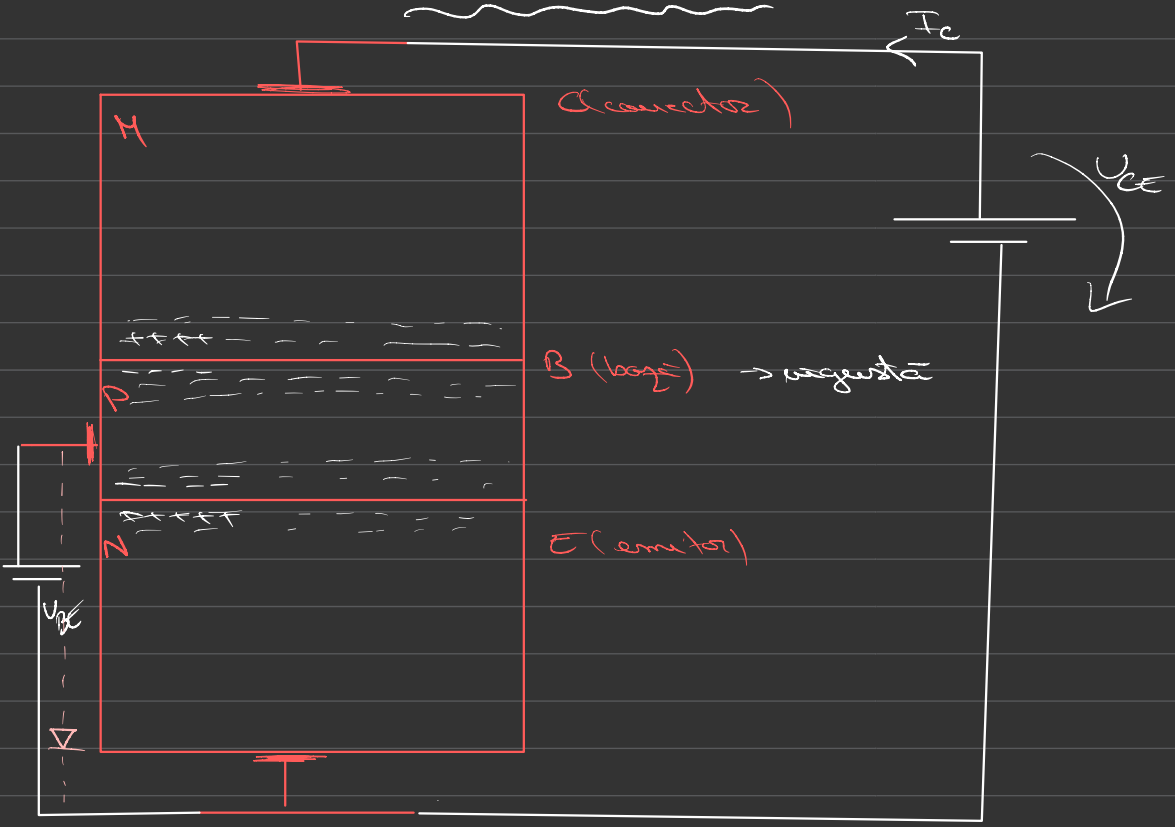
(900 MHz → 3,5 GHz / pentru telefoane)

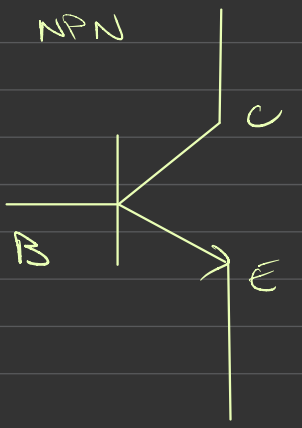


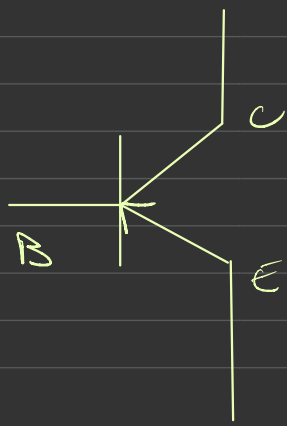
Detector fără rezistență



TRANZISTOR BIPOLAR







$$\begin{cases} U_{BE} > 0 \\ U_{CE} > U_{BE} \end{cases}$$

Regiune activă normală

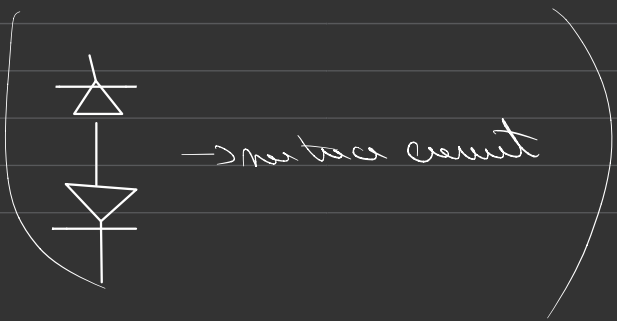
$I_C \gg I_B$

$I_E = I_B + I_C$

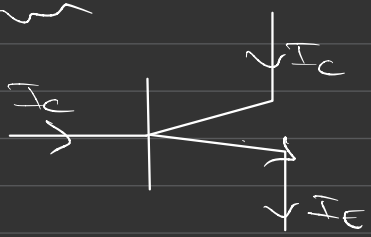
$I_B \ll I_C$

$$\boxed{I_C = \beta I_B}$$

β = constantă de fabricație (de ordinul sutelor)



PAN

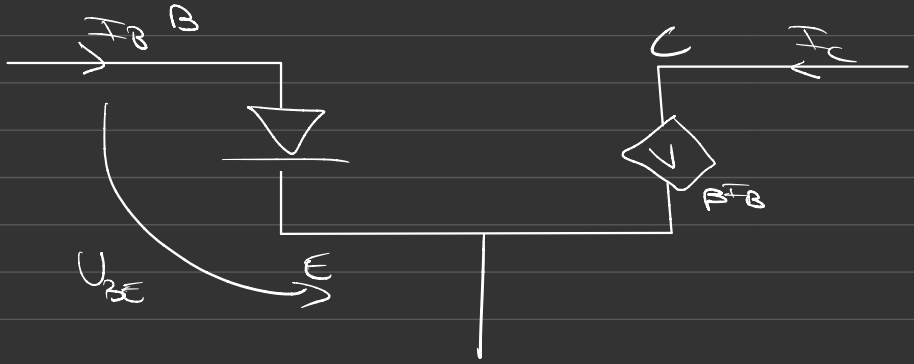


$I_C = \beta I_B$ $\quad I_B \ll I_C$

$I_E = I_C + I_B \simeq I_C$

$U_{BE} \simeq 0{,}6\ V$

MODEL SEMNAL MARE

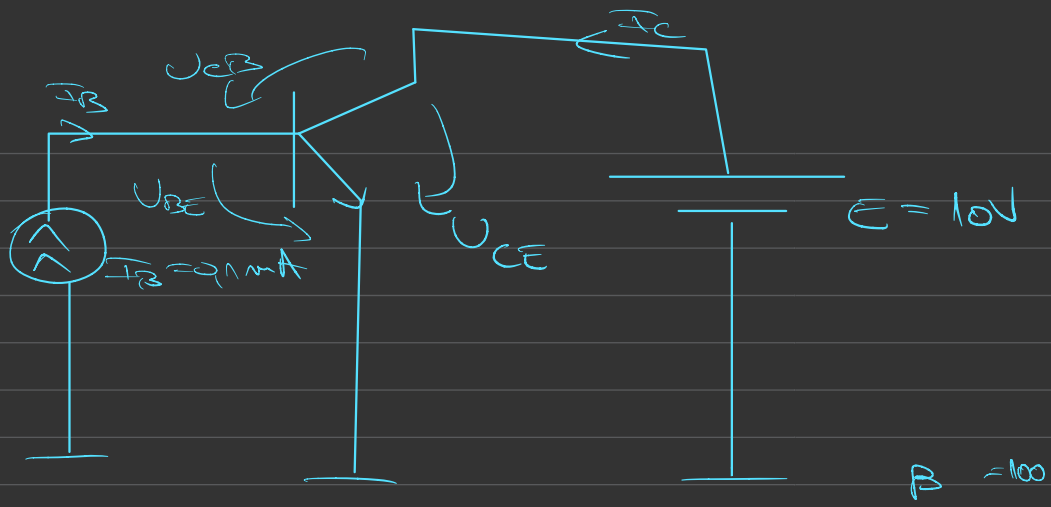


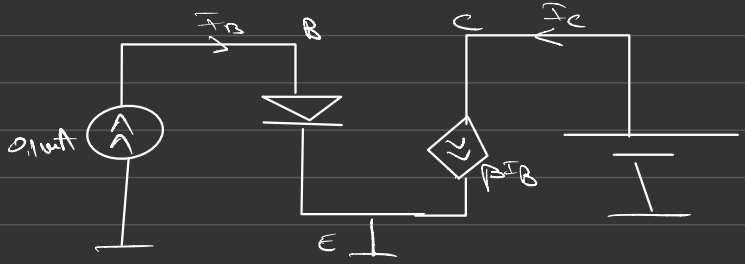
(BE) POL DIRECT

$U_{BE} = 0{,}6\ V$

(eB) POL INVERS

$U_{CB} > 0$





$I_B = 0,1 mA \Rightarrow U_{BE} = 0,6 V$

$I_C = \beta I_B = 100 \cdot 0,1 mA = 10 mA$

$U_{CE} = \mathcal{E} = 10V \Rightarrow U_{CB} = U_{CE} - U_{BE} = 9,4V$